ECON 0150 | Homework 2.3

Due: Friday, Oct. 9 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Q1. Explore the Data

The dataset `amazon_book_sales.csv` lists Amazon's 50 bestselling books each year from 2009 to 2022. Prices are in US dollars, rounded to the nearest dollar. The column `Genre` marks each book as Fiction or Non Fiction.

a) How many unique books are in the dataset? *(hint: use `.unique()` to count a book only once even if it shows up in multiple years)*

b) How many books made the bestseller list in more than one year? *(hint: `.value_counts()` counts how many times each `Name` appears)*

Q2. Price and Ratings by Genre

a) Create a scatterplot of `Price` (x-axis) vs `User Rating` (y-axis), colored by `Genre`.

b) Is the relationship between price and rating positive, negative, or unclear?

c) Does the relationship look different for Fiction and Non Fiction?

Q3. Reviews and Ratings

a) Create a scatterplot of `Reviews` (x-axis) vs `User Rating` (y-axis) for all books.

b) Is there a relationship between reviews and ratings? If so, is it positive, negative, or unclear?

c) How might you use a transformation to improve this figure?

Q4. Does the Relationship Differ by Genre?

a) Create a scatterplot of `Reviews` vs `User Rating`, colored by `Genre`. Use any transformation you suggested in Q3.c).

b) Does the relationship between reviews and ratings differ between Fiction and Non Fiction? Describe the pattern.

c) In one sentence, what hypothesis might explain the difference you observe?